AI-Driven Triage: A Graph Neural Network Approach for Prehospital Emergency Triage Patients in IoMT-Based Telemedicine Systems

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Abstract— **Improper triage, overcrowding in the emergency department (ED), and long wait times for patients are some of the problems that arise from emergency triage due to a lack of medical and human resources. In emergency departments, proper patient triage is crucial for determining the urgency and appropriateness of treatment by evaluating the severity of each patient's condition. In most cases, a human operator will use the knowledge and data gathered from patient management to perform triage tactics. Hence, it is a method with the possibility of causing problems in high-priority interconnections. Currently, typical triage systems primarily rely on human recommendations, which are susceptible to personal convictions and error. Methods that maximize data collection and remove mistakes in patient triage processing are being developed with an increasing amount of interest in utilizing artificial intelligence (AI). We develop and deploy an AI module to control the distribution of emergency triage patients in EDs. We use data from past emergency room visits to educate the medical decision-making process. Patients are properly sorted into triage groups because of data including vital signs, symptoms, and an ECG. Test results have shown that the proposed algorithm is better than the currently used state-of-the-art triage techniques. In our opinion, the methods used will help healthcare professionals forecast the severity index, which in turn will help them direct the proper patient management protocols and allocate resources.**

Keywords— AI-Driven Triage, Prehospital Emergency, GNN, IoMT, Telemedicine

I. INTRODUCTION

The need for emergency triage in medical care has significantly spread because of a large increase in emergency departments (ED) in the last couple of years [1]. In fact, the increased number of visits to emergency departments has become a cause for concern as a result of overcrowding, a rise in patient queues, and a lack of healthcare facilities and services [2]. Additionally, these challenges have caused a specific number of deaths among patients who are in critical conditions [3]. The present approach to triaging patients in the emergency unit consists mainly of physicians making observations, talking, and using devices and equipment to examine vital signs in order to evaluate the patient's level of urgency and determine the most appropriate clinical department for their treatment. As the triage workload increases, typical manual triage processes are no longer adequate. Therefore, there is an urgent need for a highly accurate and dependable triage technique or system [4]. Regarding triage methods, the two most often used ones globally are the Canadian Triage and Acuity Scale and the Manchester Triage System. In order to prioritize patients, these systems identify them based on their health indicators and symptoms. By incorporating both organized and unorganized medical information, such as the Emergency Severity Index, is another advantage of integrated models [5] and the Chinese Emergency Triage Scale [6]. However, traditional emergency department triage systems are labor-intensive and time-consuming since they rely on clinical assessment and doctors must manually perform patient triage. In addition, there may be difficulties in assigning priorities to patients according to their criticality levels if both objective and subjective variables are used to classify their levels of urgent [7].

In perspective of the significant applications of AI in healthcare, especially in ED patient triage management, we investigated the feasibility of exhibiting patients in various similarity networks utilizing three distinct Graph Neural Network designs. Current prediction models employ tabular data acquired at triage. Nevertheless, the process of discovering commonalities among individuals could be beneficial in forecasting admissions during the triage phase. In regard to this, we propose including the patient's properties into the networks across the patient entity and the variables related to it. We also suggest using edges with weights to show how similar patients are. Within a classical triage system, patients who need to be admitted to the Emergency Department are categorized into different levels of severity using established algorithms. These computations calculate an indicator of severity by examining the patient's data. The nurse can gather such information by monitoring the patient and measuring biochemical indications. This type of data is typically stored in hospital registries and electronic health records (EHRs).

These challenges highlight the need for a more accurate and effective triage mechanism that is also reliable. The latest advancements in artificial intelligence, especially in the fields of machine learning and graph neural networks, provide possible solutions to these problems. By making use of graph

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neural networks for the purpose of expressing detailed relationships and similarities among patients, we can make an AI-powered triage system that is more accurate and efficient in emergency medical care. This study investigates graph neural networks' implications for the growth of patient representation in similarity networks. All the patients are considered as nodes of a network, and they are connected through the common characteristics that they share. We suggest a new tactic that integrates patient features within these networks and uses complex GNN structures, such as Graph Convolutional Networks, Graph Attention Networks, and Graph SAGE, to sort out patients by their level of emergency. Thus, our method not only solves the existing problem of trifling frameworks but also paves the way for more rapid and accurate patient prioritization in emergencies.

II. RELATED WORKS

With the objective to improve healthcare for patients, increase survival rates, and support in critical decision-making, The medical business currently widely utilizes artificial intelligence[8]. presented here are some illustrations: Applying a Decision Tree to address the challenge of improving energy efficiency in distributed federated learning offloading and scheduling in blockchain-based networks. This involves developing a fog computing resource management model that combines hybrid load balancing and scheduling techniques specifically for critical healthcare applications. the researchers are working towards developing AI-based Diagnostic systems for equipment operating with low resources [9]. Researchers have further emphasized the use of deep learning (DL) classification models for myocardial infarction patients. The present models employ Grad-CAM technology to assess the results of the model [10]. Moreover, a machine learning-based predictive model has been created to predict sepsis episodes in ICU patients, which makes it possible to diagnose it several hours before it happens. This development can shorten a patient's life by cutting down on the time frame within which the correct medication is administered. [11]. via integrating artificial intelligence (AI) modern technology into the emergency department (ED) triage procedure, it is achievable to overcome limitations in resources while minimizing the demands on healthcare professionals while enhancing the accuracy and efficiency of patient prioritization. A number of studies have been attempted to reach this goal. Dugas et al. [12] Developed a technology-based triage system which utilized regular triage data to determine patient allocation probability and provide comprehensive findings. In contrast to the ESI, the method reduced selectivity and improved medical decision-making. Zhang et al. [13]established a predictive framework that forecasts hospitalization or referral outcomes using Convolutional Neural Networks and Natural Language Processing. This technique has an opportunity to help medical professionals make decisions and distribute the resources they have. Kim et al. did a study [14]They investigated the efficacy of different machine learning algorithms and medical evaluating systems in prioritizing suspected sepsis patients. The study concluded that machine learning algorithms, when supplemented with additional patient complaints, exhibited superior performance. In a more recent study, El-Bouri et al. [15] introduced a deep learning-based approach for forecasting the location where patients are admitted following triage in the emergency department. Boussen et al.[16] employed an unsupervised machine learning technique to group together Information related to the saturation level of oxygen and respiratory rate of COVID-19 patients. This approach successfully evaluated the seriousness of patients' conditions and forecasted the duration of their hospital stay.

While these results demonstrate AI's enormous ability for enhancing outcomes for patients in emergency department triage, further study faces numerous challenges. Despite the importance of unstructured triage data, several studies simply look at patients' vital signs. In addition, overall triage, which is the main reason patients attend, has received minimal research compared to studies that anticipate patients' urgency or the need for medical treatment or hospitalization. In order to overcome these obstacles and enhance the provision of healthcare in times of emergency, it is required to implement a more comprehensive triage strategy in the emergency department.

III. SYSTEM DESIGN AND IMPLEMENTATION

This study leverages Graph Representation Learning techniques integrated into a Graph Neural Network framework to improve the efficacy of medical emergency triage in hospitals. The approach we employ comprises of several critical stages, including data collection, the preprocessing phase the development of patient similarity networks, and the use of advanced GNN designs for patient assessment. The approach which we employ is Graph Representation Learning, which is implemented on networks Graph Neural Networks were utilized. Graph Representation Learning is a method that uses dimensionality reduction vectors for expressing graph structure data. The structure of the original graph is accurately represented by a map that is achieved by mapping a graph's nodes or subgraphs onto a vector space. GRL makes it easy to use the embedded data you get as variable inputs for tasks related to machine learning later on, so you can use graph structures for all sorts of activities including Tasks in the field include node categorization, connection prediction, and graph visualization. Several GRL methodologies have been established, however we have specifically opted for Graph Neural Networks that employ inductive techniques to handle node labels.

The system's architecture is illustrated in Figure 1. The method uses patient data to develop a network to illustrate how similar patients relate to one another, with weighted edges indicating the degree of similarity between each patient. Cosine and Manhattan distances are two popular measures for determining how similar two objects are in the present setup of a system. One of the features of the network embedding module also called graph representation learning is to use node embedding to map the network onto a hidden field. The GRL technique preserves the initially set distance between nodes while mapping them onto individual points inside a subspace. The node classification module is responsible for determining a triage level to each new patient that has to be provided [17]. Both the module for learning graph representations and the module for classifying nodes utilize the computational intelligence of Graph Neural Network approaches. We choose the techniques for Graph Representation Learning (GRL) that are capable of handling

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node labels using inductive procedures[18]. Graph Neural Networks are utilized for node embedding due to their two primary advantages: firstly, they consider node-related data, such as node features; secondly, they possess an inherent inductive property, preventing an additional calculation of entire embedding in the event of any modifications to the graph, such as node or edge insertion or removal.[19]. The present approach utilizes three cutting-edge techniques: Graph Convolutional Networks [20], GAT [21], and Graph Sage [22],[23], which, for overall, outperform the previous methodologies. For effective deployment, GNN-based systems must interact with current EHR systems and facilitate clinical decision-making while preserving human judgment. AI recommendations can enhance existing triage procedures by delivering real-time updates derived from ongoing patient data. Effective interface design and clinician training are essential for guaranteeing seamless integration.

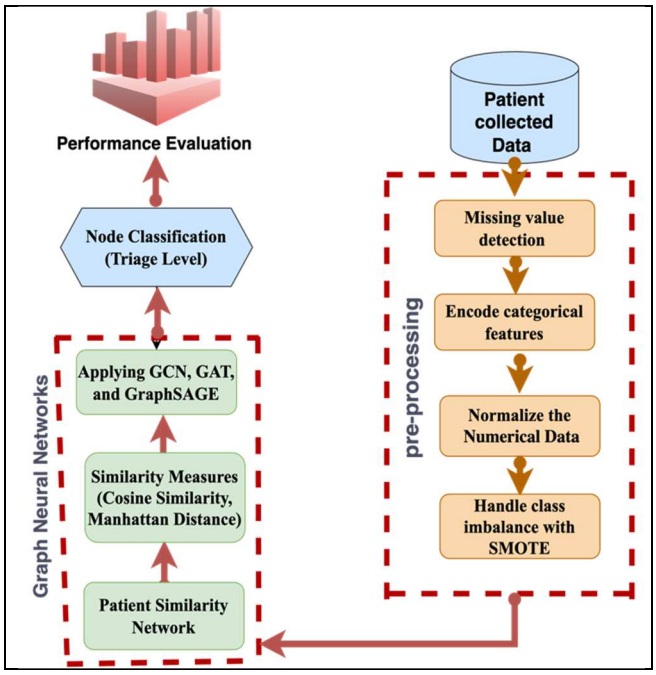


Fig. 1. Proposed framework

A. Data Collection and Preprocessing

The dataset used consists of 50,000 instances with several features related to patient data and triage levels. The target variable is 'TRIAGE', which denotes the triage category assigned to each patient. The four text (non-sensory) elements, the sensors, the BP, SpO2, and the ECG are derived from [24]. Furthermore, the suggested approach takes into account the updated dataset that includes additional textual (non-sensory) elements like headache and left-hand pain, as well as new sensor information like temperature [25]. Table 2 displays all of the data that was prepared in accordance with medical criteria. The possibility of differences in values was evaluated by looking at the patient's diagnoses and symptoms. Notably, 50,000 sets of simulated sequences were used to simulate 50,000 people with two types of multiple chronic illnesses (MCDs), hypertension, and chronic heart disease. Patient queries are estimated using information collected by the hospital's data center that manages urban healthcare. Following an evaluation by specialists in chronic illnesses, each patient's symptoms were divided into five severity-based categories. (risk, urgent, sick, cold case, and normal) according to established medical protocols[26]. While conducting the research, the data underwent a cleaning process to eliminate any missing or inconsistent items, so ensuring the reliability of the future study. Subsequently, the data underwent standardization to a uniform scale, a critical step for optimizing the performance of machine learning models. Z-score normalization was utilized for continuous variables, such as vital signs. The categorical data, such as patient symptoms, were transformed into a numerical format using one-hot encoding. This was done to make them appropriate for input into the GNN models.

B. Graph Convolutional Networks (GCNs)

Graph Convolutional Networks [27] are neural network architectures that have been designed to accept input arranged as graphs. These networks are extremely useful in conditions with associated data points, for example, systems of interactions between people, chemical composition, or networks of communication. GCNs analyze data on nodes and their connections using the graph structure, especially as opposed to standard convolutional neural networks working with grid-like input like pictures [27]. Convolution operations take place immediately on the graph, enabling information to be aggregated from surrounding nodes. The above procedure records the properties of each node as well as the complicated relations between them. GCNs are well-suited for Applications such as node classification and link prediction and graph classification because of their ability to manage irregular graph structures.

C. Graph Attention Networks (GATs)

Graph Attention Networks [28] Use an attention strategy to automatically assess the value of each nearby entity's attributes, enabling advanced and context relevant information combinations. This method builds on the basis of standard graph convolutional networks, which use the characteristics of nearby nodes either in preparation or equally. Graph Attention Networks utilize an attention mechanism to allocate various weights to nodes that are more relevant inside a specific neighborhood. While dealing with complicated network structures, where neighboring nodes' relative value could change significantly, this comes in extremely helpful. Using this approach, features may be represented and obtained more efficiently.

D. GraphSAGE

Graph SAGE [22] is a technological method designed to generate embeddings of nodes for massive graphs. Unlike typical graph-based techniques, this approach gathers and integrates data from a node's immediate surroundings, seeking to effectively oversee extensive networks of interconnected nodes. It can aggregate data using various methods and effectively capture a wide range of neighborhood structures. Graph SAGE can do inductive learning and is exceptionally efficient in Applications involving massive amounts of dynamic graph data, including categorization, prediction, and recommendation systems.

IV. Results and Discussion

This research investigated the performance of three complex Graph Neural Network structures: Graph Convolutional

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Networks, Graph Attention Networks, and Graph SAGE. These models were utilized for classifying emergency triage patients based on a variety of input factors, including vital signs and other clinical data. The GCN model achieved an accuracy rate of 77.50%. While it accurately captured the local structure of the patient similarity graph, it lacked the precision of GATv2Conv, which surpassed GCNs having an accuracy score 82.00%. The use of the attraction method allowed the framework to assign variable degrees of significance to different neighbors, improving classification accuracy. Nevertheless, The Graph SAGE model exceeded the other models, with an accuracy of 95.80%. Graph Sage's capacity to generalize to novel, unseen data and its inductive learning methodology rendered it the most resilient model in our experiments.

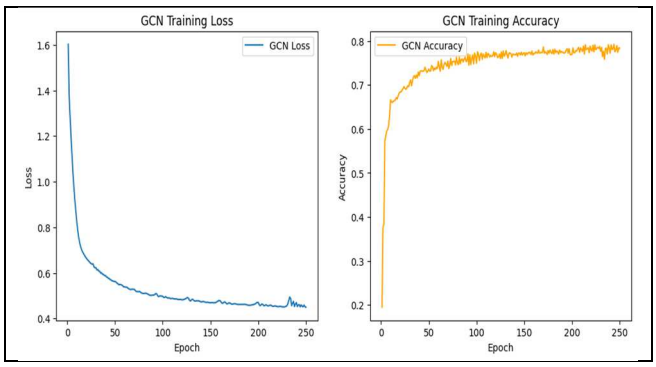


Fig. 2. GCN Trining Loos and Accuracy

Figure 2 demonstrates the efficacy of the GCN model in acquiring knowledge from the data. The consistent decline in loss and the corresponding rise in accuracy demonstrate that the model is functioning effectively during training. Nevertheless, the ultimate accuracy of approximately 77.50% suggests that although GCNs possess capability, they may not completely comprehend the intricacy of the task in comparison to more sophisticated models such as GATv2Conv or Graph SAGE

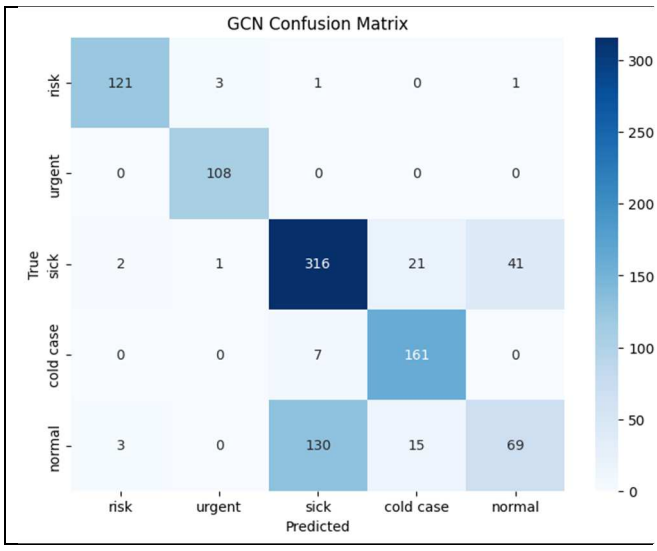


Fig. 3. GCN Confusion Matrix

The confusion matrix of the GCN, as shown in Figure 3, offers a comprehensive assessment of the model's effectiveness in various triage areas. Although the model demonstrates strong performance overall, it can be further enhanced in terms of reducing misclassifications, notably in distinguishing between "sick" and "normal" categories. However, it excels at accurately recognizing "urgent" and "sick" patients. This research is crucial for comprehending the advantages and constraints of the GCN model in the emergency triage scenario and aids in directing future enhancements to the model.

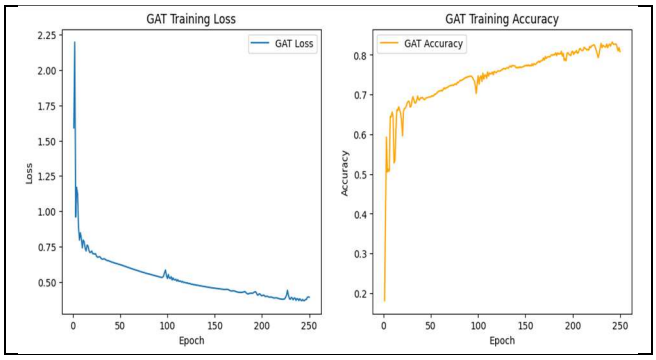


Fig. 4. GAT Trining Loos and Accuracy

The training curves in Figure 4 show the GAT algorithm's ability to discover the underlying patterns in the data, which is a major reason for the final performance being so good. The loss decreases steadily while accuracy rises continuously and reaches its maximum stable value of about 80 percent. It shows that the GAT algorithm has been well developed to accurately predict, but the AUC of certain categories can still be improved. This chart provides valuable learning insights into the GAT model, enabling it to be assessed for its possible effectiveness in a real-world emergency triage setting. The model's steep initial learning curve, followed by a steady state of accuracy, indicates that it might be an asset in time-sensitive scenarios.

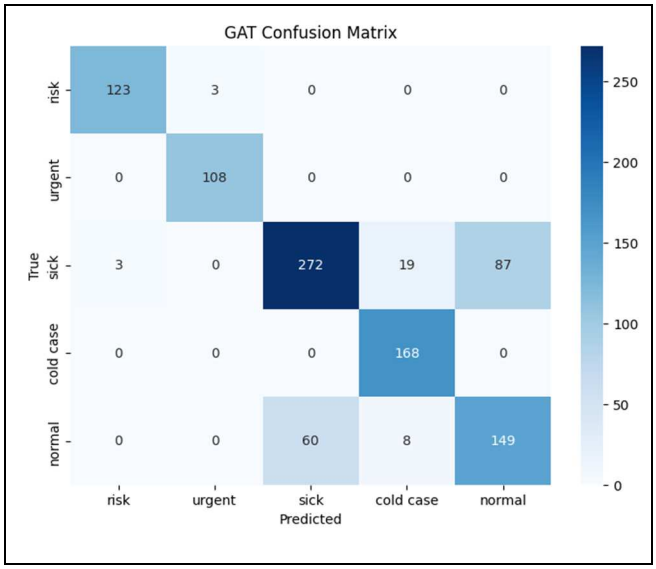


Fig. 5. GAT Confusion Matrix

Figure 5, on the GAT's confusion matrix, gives a clear depiction of the accuracy in classifying the different triage categories according to the model, thus identifying its strengths and weaknesses. The algorithm not only efficiently determines the patients who need urgent care and those with chronic illnesses, but also merges these two categories into

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one with only a minimal number of misclassification problems. Nevertheless, the model confronts hurdles in distinguishing the "sick" from the "normal" patients, which is significant gap that needs to be overcome to boost its success rate in differentiating these quite similar categories.

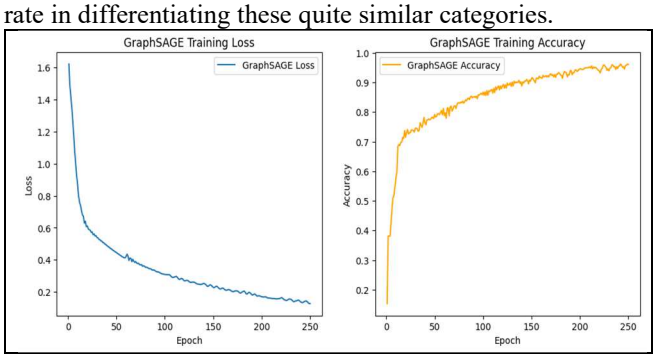


Fig. 6. GraphSAGE Trining Loos and Accuracy

Figure 6, demonstrating the model's learning curve, shows unwavering consistency in the model's learning capacity even when the input data is limited. The sharp slope of declining loss and ensuing rescaled accuracy highlights the model's strong ability to easily recognize and accurately scan out the right classes. Graph SAGE positioned itself as an impressively accurate predictor on the training set, thus becoming the most credible model for the task. According to the study, it is the most consistent and precise model among the test models and therefore provides a complementary solution for tasks that require precision and generalization, such as emergency triage in hospitals.

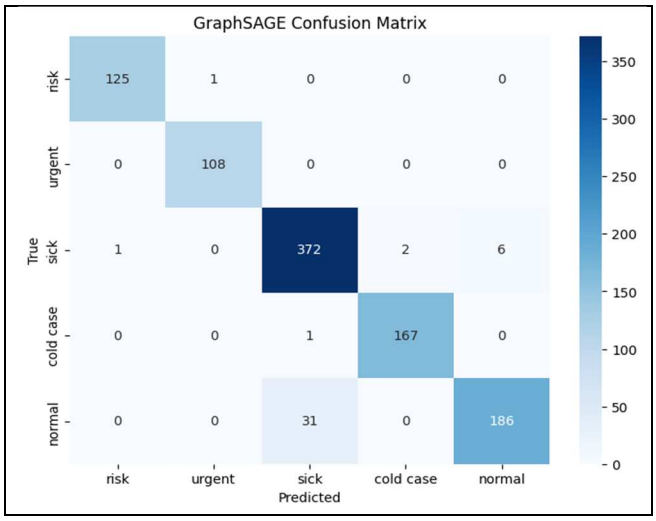


Fig. 7. GraphSAGE Confusion Matrix

The confusion matrix shown in Figure 7 highlights the accuracy of Graph SAGE in classifying patients into the correct triage groups. In terms of performance, this model is ahead of the others, for instance, GCN and GAT. Due to the low rate of misclassifications, that is especially true in groups such as "risk" and "urgent", Graph SAGE is a good option for the tasks that are essential for reliable prioritization. This analysis shows the resistance and accuracy that Graph SAGE possesses, proving that it is a suitable model to improve the triage process in emergency cases. The model's generalization and its ability to effectively predict outcomes in various categories have made it relevant in the healthcare environment.

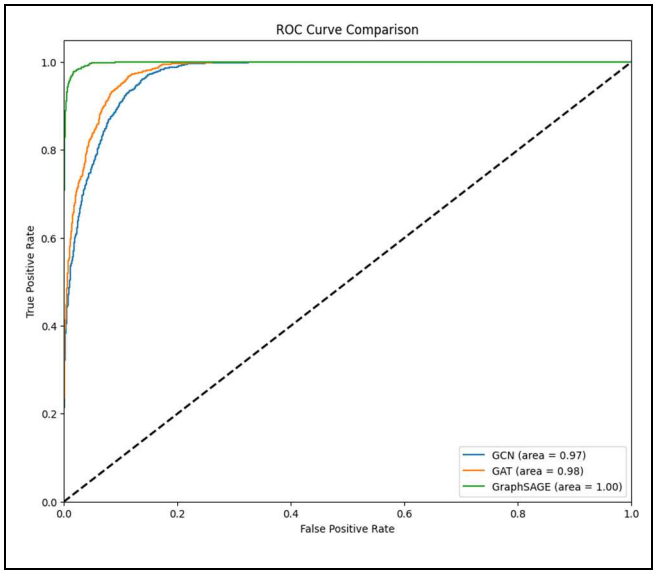


Fig. 8. ROC Curve For All Techniques

In spite of the fact that the three models show a good performance, as it is evident from Figure 8, the GCN is relatively inefficient compared to GAT, and GraphSAGE is the best among the three. The attention technique GAT is different from Graph Convolutional Network (GCN) in the following way: it helps the algorithm concentrate on the crucial features; thus, the overall performance is achieved. Nevertheless, GraphSAGE is a noticeable improvement in accuracy, which is most probably because of its superior aggregation methods and inductively learned capabilities. The ROC curve analysis highlights the exceptional performance of the GraphSAGE model in this study, with its superb AUC being very close to the perfect classification of classes. Although the GAT model performs better than the GCN model, the GraphSAGE model eventually outperforms it. This research draws the conclusion that GraphSAGE is the ideal model for this purpose, being very accurate and reliable in the classification of emergency triage patients.

The use of GNN-based triage systems in emergency departments encounters numerous challenges. This encompasses substantial computational requirements for real-time processing, the integration of AI with current medical systems, and the assurance of data protection. Furthermore, the interpretability of AI models is essential for physician adoption, as they must comprehend and trust AI-generated suggestions. Effective training and ongoing system validation are essential for effective implementation. The real-time application of AI in emergency departments presents problems like processing speed and the adaptation of models to swiftly changing settings. Hospitals must invest in infrastructure that supports real-time AI predictions, while consistently monitoring and upgrading the model to maintain stability amid varying workloads.

V. Conclusion

This research recommends applying graph neural networks to strengthen emergency triage processes in medical facilities. Our aim was to ensure the proper classification of the patients in the emergency departments in the case of errors and over-

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triage using graph convolutional networks, graph attention networks, and graphSAGE, the most recent GNN architectures. Our strategy was to create networks of patients' similarities through the use of medical data that presented a much more exact picture of patients' connections and could lead to more accurate triaging. The findings of the study show that GraphSAGE performed better than the two other GNN models, with an accuracy of 95.80%. GraphSAGE's extraordinary performance is feasible due to its ability to apply inductive learning techniques. Further, the latter allows the model to swiftly comprehend new and hitherto unseen aspects. Such adaptability is particularly essential in such volatile and high-stakes settings as emergency departments (EDs). The model's capacity to get and analyze the data from the context around a patient played a key role in attaining correct and reliable classifications of all triage levels. Findings suggest that a graph neural network, particularly GraphSAGE, has the potential to entirely change the triage process in EDs. Therefore, the artificial intelligence (AI) solutions that will help in the classification of patients will lead to a shorter waiting time, improved health outcomes, and the use of resources more efficiently. A rapid and correct classification of patients based on the degree of urgency of their health conditions is remarkably important, as often the outcome arises with time as a decisive factor. However, the experiment was mostly based on the assumption of ideal or unskilled residents; therefore, some issues still need to be addressed. Further studies should emphasize incorporating unstructured data, such as medical records and imaging data, into GNN models for the enhancement of the accuracy of predictions. It is, moreover, of utmost importance to determine the GNN models' robustness to larger patient data and their integration with the already existing hospital information system. This also concerns the evaluation of the computational efficacy of the frameworks during real-time applications and ensuring their effective combination with the existing healthcare systems. Subsequent study will investigate the incorporation of unstructured data (e.g., medical photographs) into GNN models and the expansion of these models for application in other healthcare settings. Furthermore, enhancing real-time processing speed and guaranteeing the system's adaptability to diverse healthcare environments are essential subsequent measures for broader implementation.

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